EXISTENCE OF DIRECT SUMS IN THE DERIVED CATEGORY OF QUASI-COHERENT MODULES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file perfect.tex; no statement or proof step was rewritten, reordered or removed. The counted span is the author environment and its own proof environment. References inside the span are handed to the assembler as navigation stubs that carry the canonical tag of the cited tag only; whether each of them is load-bearing is recorded on the stub itself.

Allowed standard prerequisites (editorial).

- The derived category D(O_X) has direct sums and they are computed termwise on representatives. invoked at proof lines 303-305. Provides the candidate direct sum in D(O_X).
- Taking direct sums is exact in any Grothendieck abelian category, so cohomology of a termwise direct sum is the direct sum of the cohomology sheaves. used at proof lines 306-308. Identifies the cohomology sheaves of the direct sum.
- A direct sum of quasi-coherent sheaves is quasi-coherent. invoked at proof lines 309-310. Concludes the direct sum lies in D_QCoh.

Source reference. Original lemma 07D9: sources/stacks-chow/tags.txt, source lines 9404–9404. Cited at proof lines 303-304 for the termwise direct sums in D(O_X); the quasi-coherence conclusion is inside the counted span.

Source reference. Original section 01LA: sources/stacks-chow/tags.txt, source lines 2079–2079. Cited at proof lines 309-310 for the quasi-coherence of direct sums; the conclusion is inside the counted span.

Lemma 0.2. *Let X be a scheme. Then $D_{QCoh}(\mathcal{O}_X)$ has direct sums.*

Proof. By Injectives, Lemma 07D9 the derived category $D(\mathcal{O}_X)$ has direct sums and they are computed by taking termwise direct sums of any representatives. Thus it is clear that the cohomology sheaf of a direct sum is the direct sum of the cohomology sheaves as taking direct sums is an exact functor (in any Grothendieck abelian category). The lemma follows as the direct sum of quasi-coherent sheaves is quasi-coherent, see Schemes, Section 01LA. □

Date: stacks-project:perfect.tex:local-snapshot-20261012; GNU FDL 1.2 or later.